Generation of stable cell lines using DNA transposases

DNA transposons are mobile genetic elements that integrate transgenes into chromosomal DNA via a cut-and-paste mechanism. They are mobilized by transient expression of a transposase, which excises the donor sequence and inserts it into target sites through strand transfer reactions—without causing double-strand breaks at the insertion site.

Different transposase systems have been developed to support stable transgene integration in mammalian cells. These systems vary in integration efficiency, sequence preference, and genomic stability.

The Sleeping Beauty (SB) system (Ivics et al., 1997), including hyperactive versions such as hySB100X, integrates at near-random sites but has relatively low transposition efficiency and a tendency for local hopping on the X chromosome.

The PiggyBac (PB) system (Ding et al., 2005), by contrast, is ten- to hundred-fold more efficient, with no evidence of local hopping. It shows a preference for intragenic regions such as transcription start sites, making it well-suited for insertional mutagenesis.

Risk assessment

- – Work with human-derived material or transgenic cell lines (BSL-2)
- ▷ Wear gloves, safety glasses, lab coat
- ☐ Dispose waste after inactivation as REGULATED MEDICAL WASTE



Reviewed: Nov 12, 2023

Procedures

Transfection of host cells

- ☐ 6-well plate
- ☐ Transfection reagent
- ☐ Transposase helper plasmid
- ☐ Transfer plasmid

(1.) Seed cells to reach 70% confluence (50 000 cells per well in a 6-well plate) at the time of transfection.

(2.) Prepare a mix of the transposase and transfer plasmids at a sub-stoichiometric ratio.

Plasmid		Per well	
Transposase plasmid	Super PiggyBac transposase (SBI)	0.05 pmol	0.32 µg
Transfer plasmid	Custom	0.20 pmol	1.00 µg

> ***Note:*** Transposase plasmids often encode a transiently expressed fluorescent protein to track transfection efficiency. The number of integrated copies, however, depends on transposase activity, not transfection efficiency.

(3.) Transfect the cells using a non-viral method such as Lipofectamine® SOP 0012.

(4.) After 24 h, replace the medium.

(5.) Begin selection or screening for stably transfected cells according to SOP 0019.

> ***Hint:*** Selection can begin as early as 24 h post-transfection, but may benefit from a 48 h recovery window in sensitive lines.

> ***Critical:*** The Limited Use Research License (LURL) of the piggyBac transposase (System Biosciences/Demeetra AgBio) applies to all derived cell lines, persists until those lines are destroyed, and excludes any transfer to, use by, or contracted work for industry or commercial third parties.

Materials

Transfection of host cells

6-well plate

Transfection reagent, non-viral

Hint: see SOP 0012 for selection of a transfection method.

Transposase helper plasmid

- **Super PiggyBac Transposase Expression Vector**, System Biosciences, PB210PA

Hint: The piggyBac transposase plasmid may not be redistributed; customers cannot propagate or re-clone the plasmid and must repurchase from System Biosciences (the academic distributor licensed by Demeetra AgBio) when stock is exhausted.

Transfer plasmid

- *e.g.* **PB-CAG:MCS-PGK:HygroR**, constitutive CAG, hygromycin selection, RRID Addgene_48753
- *e.g.* **PB-CMV:MCS-PGK:PuroR**, constitutive CMV, puromycin selection, RRID Addgene_219794
- *e.g.* **PB-TRE:EGFP-EF1a:rtTA-T2A-PuroR**, Tet-On inducible (rtTA), puromycin selection, RRID Addgene_104454

Hint: Addgene distributes piggyBac transfer plasmids under a Limited Use Label License (LULL) for academic non-profit research. No transfer to third parties, including institutional core facilities. Two-year license term; re-request from Addgene to renew. Re-cloning the ITRs into a different backbone or synthesizing them de novo falls under Demeetra AgBio patents requires separate licensing.

List of references

S. Ding, X. Wu, G. Li, M. Han, Y. Zhuang, and T. Xu, *Cell* **122**(3), 473—483 (2005).

Z. Ivics, P. Hackett, R. Plasterk, and Z. Izsvák, *Cell* **91**(4), 501—510 (1997).

Change log

2023-11-12	Benjamin C. Buchmuller	Adaptation as SOP, synthesizing the Sleeping Beauty and piggyBac transposon literature cited in the body; donor-to-helper plasmid amounts from Shany Koren-Hauer (Tom Muir lab).

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